BSc (Hons) Computer Science
BSc (Hons) Cyber Security
Module Code: QH0305 Module Title: Problem Solving
Assessment Sheet 4

Instructions:

This is one of the eight assessment tasks which will contribute to the overall mark. You will need to complete the tasks as outlined below and then document them in a Word document file. As a minimum, you should provide screenshots of the following:

- Your code
- The output that your code generates

In instances where your code could generate different outputs depending on what values are given, you should provide multiple screenshots of the console screen, showing the different outputs in order to demonstrate that the code works correctly.

Your Word document should have appropriate headings to ensure that each task can easily be identified alongside the rest of your work.

This assessment will focus on loops, conditional logic, and calendar formatting.

You must attempt all tasks on this sheet to achieve a higher grade. For example, if you want to gain marks between 70 - 100, you must complete all other grades first and add them to your portfolio with screenshots.

A zip folder with all Grade codes (in .c or .txt format) must be submitted alongside the portfolio (MS Word file). This is in addition to the code presented as screenshots.

Assessment Task 4: Calendar Printer

Throughout this task, you will create calendar-printing scripts, starting from very simple and incorrect (for the purpose of printing off the structure), to correct calendar for given month/year.

To achieve Grade D (Between 40-49) (Basic Calendar Grid)

Requirements:

Write a program that prints a single-month calendar for the year 2025. The program should prompt the user to enter:

- A number corresponding to a month (1–12)
- The weekday on which the month starts (1 = Monday, 7 = Sunday)

Print the calendar in a structured grid reflecting the starting weekday and a fixed number of days of your choice (e.g., 28, 30, or 31) – the aim here is layout, not real month lengths. Include screenshots of different starting-weekday cases

Note: *This is not a "real" calendar at this stage! User decides if 1st of "their month" is a Monday or a Thursday or a Sunday etc. The task tests the ability to print a calendar for any made-up or real month, based on user input.*

You need to:

- Prompt the user for **month number** and **starting weekday**.
- Use a **fixed number of days** (e.g. 30) to render the grid.
- Show neat alignment for weeks (Mon–Sun headers, day numbers beneath)

Example Interaction:

If the user inputs Month as 7 (July) and starting Weekday as 5 (Friday), the expected outcome of the program could print as below (with fixed number of days chosen as 28):

July						
S	M	T	W	T	F	S
					1	2
3	4	5	6	7	8	9
10	11	12	13	14	15	16
17	18	19	20	21	22	23
24	25	26	27	28		

Note: July has 31 days, but the aim of the tasks is to print the calendar in accordance with user's input and not a "real" calendar.

To achieve Grade C (Between 50-59) (Correct Month Length and Validation) Requirements:

Complete all previous steps, then:

- **Determine the correct number of days** for the selected month in 2025, using an appropriate logical structure (store 31 for January, 28 for February etc.)
- Print the calendar using the **correct length** for that month
- **Count and display** the number of Saturdays and Sundays
- Add **basic input validation** for month and weekday values

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To achieve Grade B (Between 60-69) (Start Date Calculation) Requirements:

Complete all previous steps, then stop asking the user for the starting weekday and calculate it automatically for 2025.

- Compute the weekday for the 1st of the selected month in 2025 using modular arithmetic/date logic (e.g., accumulate days from 1 Jan 2025 (Wednesday) to the target month and take modulo 7)
- Display the derived weekday and render the calendar accordingly
- In your portfolio, include a detailed explanation of your calculation approach.

To achieve Grade A (Between 70-100) (Calendar Navigation)

Complete all previous steps, then add following functionality:

- Prompt the user to enter a specific date (e.g., 15) and highlight that date in the printed calendar (e.g., with asterisks, brackets, or another symbol)
- Provide simple navigation: after printing the chosen month, offer to print the previous or next month of 2025 (menu choice). Keep the automatic first-day calculation for the navigated month(s)
- Ensure robust input validation for all user inputs (invalid day outside the month, invalid menu choice, etc.)
- Provide detailed explanation of the logic used in this task and describe any problems you have encountered

Assignment Preparation Guidelines

- All components of the assignment report must be Word-processed **(handwritten text or hand drawn diagrams are not acceptable)**, font size must be within the range of 11 point to 14 point including the headings, body text and any texts within diagrams.
- Standard and commonly used fonts such as Times New Roman, Arial or Calibri should be used.
- All figures, graphs and tables must be numbered and labelled with short explanations.
- Material from external sources must be properly acknowledged and cited within the text using the Harvard referencing system.
- All components of the assignment (text, diagrams, code etc.) and all assessment sheets must be submitted in one Word file.
- The report should be logically structured, the core of the report may start by defining the problem / requirements, followed by the proposed solution including a detailed discussion, analysis and evaluation, leading to the implementation and testing stage, finally a conclusion and personal reflection on learning.
- Screenshots without description / discussion are not suitable as they do not express your understanding or support your work adequately.

Submission instructions

- This is a portfolio assignment with eight tasks in total. Each task will be completed and saved in the portfolio. Once the portfolio is completed, it should be submitted on Turnitin. The submission link to Turnitin can be found under the “Assessment Tab” in your module section in the SOL VLE.

- Please note file size limitation might apply. Your report must be under 250MB.

- The source code (in .c or .txt format) for each task should be **zipped** and complete single ZIP file **submitted** to the VLE alongside the report.

- The Assignment Brief can be found under the “Assessment Tab” in your module section in the SOL VLE.

- **Refer to the Assignment Brief** to find the links to Late Submissions, Extenuating Circumstances, Academic Misconduct, Ethics Policy, Grade marking and Guidance for online submission through Solent Online Learning (SOL).